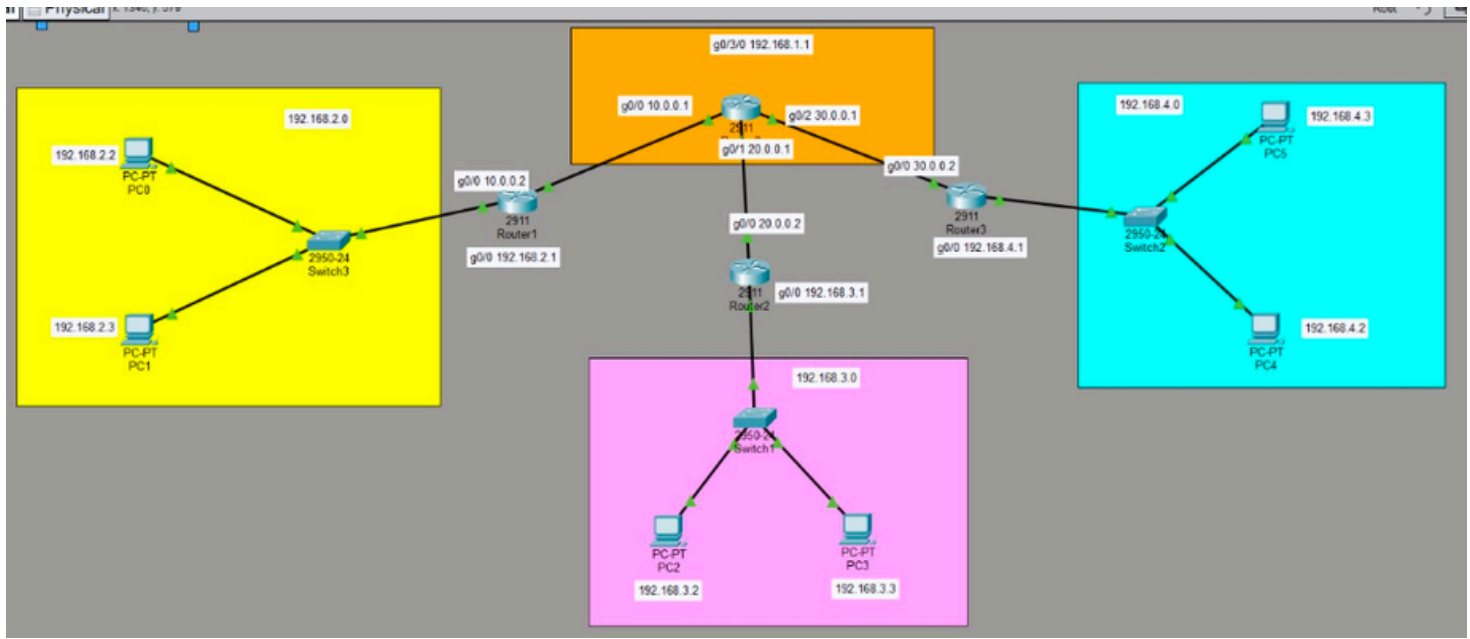


Practical 12- Mini Project

Branch Office Network using EIGRP(Enhanced Interior Gateway Routing Protocol)



Step 1:-Add devices and Create network connection

Devices:-

1.  4 Routers (2911)
2.  3 Switches (2960)
3.  6 PCs

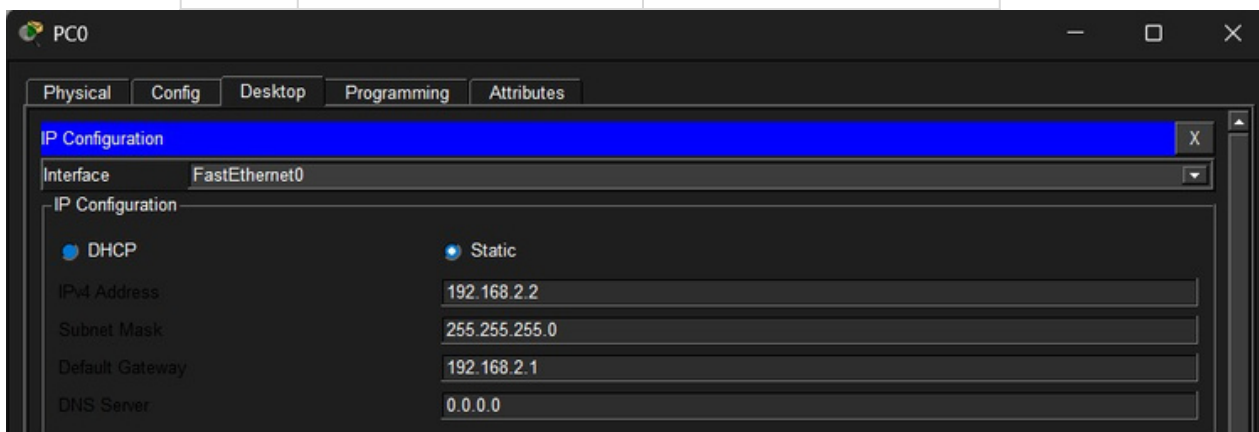
Network Connection:-

1. PC to Switch:-Use Copper Straight-Through Cable
2. Switch to Router:-Use Copper Straight-Through Cable
3. Router to Router:-Use Copper Straight-Through Cable

Step 2:-Configure IP Address on PCs

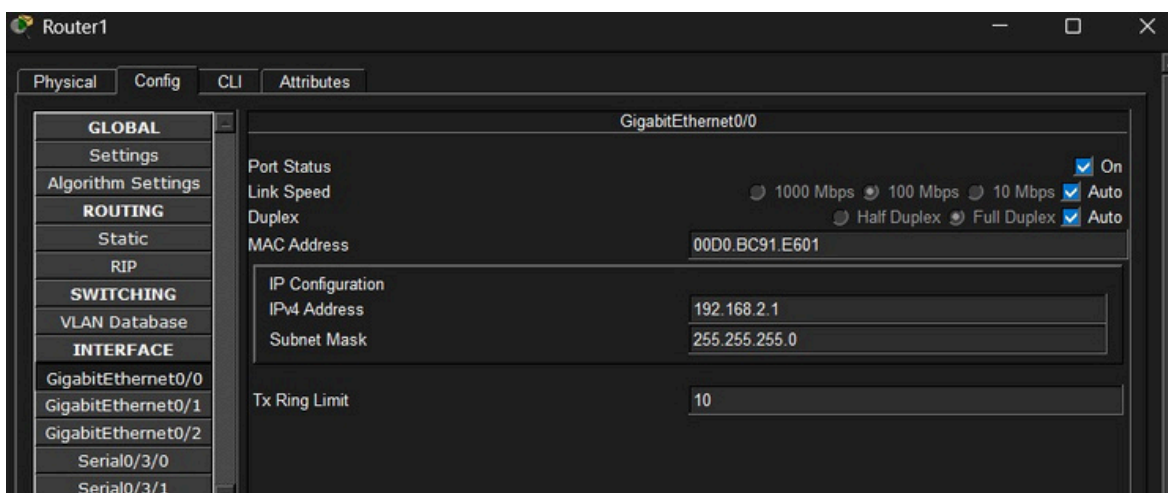
Give Ip address to all PCs

PC	IP Address	Gateway
PC0	192.168.2.2	192.168.2.1
PC1	192.168.2.3	192.168.2.1
PC2	192.168.3.2	192.168.3.1
PC3	192.168.3.3	192.168.3.1
PC4	192.168.4.2	192.168.4.1
PC5	192.168.4.3	192.168.4.1



Step 3:-Configure Router Interfaces

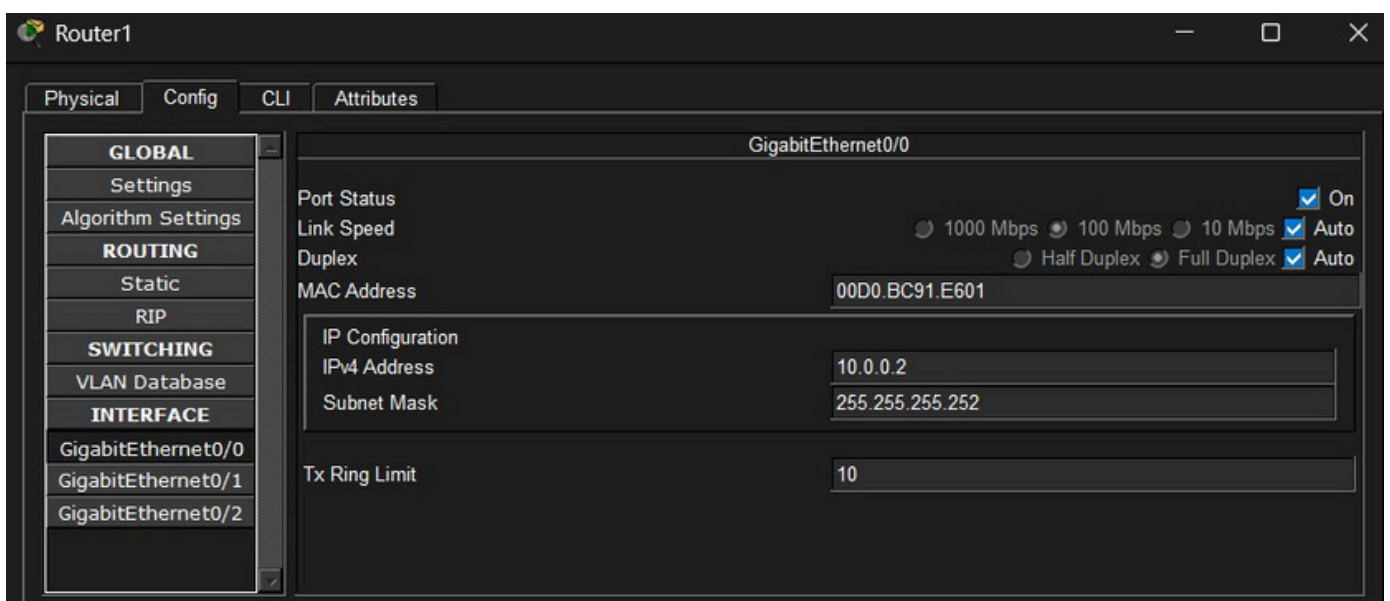
- Click on Router (e.g., Router1)
- Go to Config tab
- Click on GigabitEthernet0/0
- Enter:- IP Address & Subnet Mask
- Turn Port Status → ON



G0/1	Ip address	Subnet mask
Router 1	192.168.2.1	255.255.255.0
Router 2	192.168.3.1	255.255.255.0
Router 3	192.168.4.1	255.255.255.0

Step 4:-Configure Router to Router

- Click Router → Config
- Ethernet Connection
- Enter:
- IP Address: 10.0.0.1
- Subnet Mask: 255.255.255.252
- Turn Port Status → ON



Router 0:-

	Ip address	Subnet mask
G0/0(Router 1)	10.0.0.1	255.255.255.252
G0/1(Router 2)	20.0.0.1	255.255.255.252
G0/2(Router 3)	30.0.0.1	255.255.255.252

Router 1:-

	Ip address	Subnet mask
G0/0(Router 0)	10.0.0.2	255.255.255.252

Router 2:-

	Ip address	Subnet mask
G0/0(Router 0)	20.0.0.2	255.255.255.252

Router 3:-

	Ip address	Subnet mask
G0/0(Router 0)	30.0.0.2	255.255.255.252

Step 5:-Configure EIGRP on All Routers

👉 **EIGRP is used to automatically share routes between routers**

Router->clf

- **Enter router config mode**
- **Start EIGRP process**
- **Advertise networks**
- **Exit configuration mode**

**Similarly,do for all router ,router use only connected near by network
given example of router 0**

```
(config-if)#router eigrp 10
(config-router)#network 192.168.2.0
(config-router)#network 192.168.1.0
(config-router)#network 10.0.0.0
(config-router)#no auto-summary
(config-router)#exit
(config)#exit
```

◆ Step 6:-Verify Routing

Check routing table:-show ip route

☞ You should see routes with D (EIGRP)
similary,do for all other routers

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
20.0.0.0/30 is subnetted, 1 subnets
D       20.0.0.0/30 [90/3072] via 10.0.0.1, 00:10:53, GigabitEthernet0/0
30.0.0.0/30 is subnetted, 1 subnets
D       30.0.0.0/30 [90/3072] via 10.0.0.1, 00:10:35, GigabitEthernet0/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
D       192.168.3.0/24 [90/28672] via 10.0.0.1, 00:10:52, GigabitEthernet0/0
D       192.168.4.0/24 [90/28672] via 10.0.0.1, 00:10:34, GigabitEthernet0/0
--More--
```

step 7:- check connectivity

PC ->command prompt

Use Ping command

If configuration is correct->Reply received

Example:- PC 4 TO PC 3

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=125
Reply from 192.168.2.3: bytes=32 time<1ms TTL=125
Reply from 192.168.2.3: bytes=32 time<1ms TTL=125
Reply from 192.168.2.3: bytes=32 time=11ms TTL=125

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms

C:\>
```

